EXERCISE 4

A Non-Vanishing Wronskian Gives the General Solution

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Fundamental Set • General Solution • Uniqueness

STATEMENT

> Show that if two solutions (2) of (1) have a Wronskian that does not vanish in $[a, b]$, then (3) gives the general solution of the system.

SETUP AND HYPOTHESES

System (1) with coefficients continuous on $[a, b]$ (as in Exercise 3). (x_1, y_1) and (x_2, y_2) are solutions with $W(t) = x_1 y_2 - x_2 y_1 \neq 0$ for all $t \in [a, b]$. To show: for every solution (x, y) of (1) there are constants c_1, c_2 with $x = c_1 x_1 + c_2 x_2$ and $y = c_1 y_1 + c_2 y_2$ on $[a, b]$. (By Exercise 2 every combination is a solution, so this shows that (3) describes all of them.)

FORMULATION

Tools: (a) uniqueness for linear systems with continuous coefficients: two solutions with the same value at one instant $t_0 \in [a, b]$ coincide on all of $[a, b]$. Robinson Thm 25.2 gives this locally (near t_0); the extension to $[a, b]$ is a continuation argument: the set S of instants where the two solutions agree contains t_0, is closed (the solutions are continuous) and is open in $[a, b]$ (at each point of S the local uniqueness applies, since the coefficients are continuous), and an interval is connected, so $S = [a, b]$; (b) Cramer's rule: a 2×2 linear system with non-zero determinant has exactly one solution. Plan: match the given solution at one instant with a combination, then invoke uniqueness.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

The notes: for $n = 2$ a fundamental set is a pair of solutions with $W(\mathbf{x}_1, \mathbf{x}_2) \neq 0$, and every solution is a combination of it. This exercise proves that statement. Matching the data at t_0 is the 2×2 system with determinant $W(t_0)$, as in the second-order example of the notes (p. 5), and uniqueness does the rest.

DERIVATION

Step 1: choose the constants at one instant

Let (x, y) be any solution and fix $t_0 \in [a, b]$. We look for c_1, c_2 such that the combination equals (x, y) at t_0:

$$\begin{pmatrix} x_1(t_0) & x_2(t_0) \\ y_1(t_0) & y_2(t_0) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} x(t_0) \\ y(t_0) \end{pmatrix}$$

The determinant of the coefficient matrix is $W(t_0) \neq 0$ by hypothesis, so there is exactly one solution (Cramer):

$$c_1 = \frac{x(t_0)y_2(t_0) - x_2(t_0)y(t_0)}{W(t_0)}, \qquad c_2 = \frac{x_1(t_0)y(t_0) - x(t_0)y_1(t_0)}{W(t_0)}$$

Step 2: uniqueness

Define $\tilde{x} = c_1x_1 + c_2x_2$, $\tilde{y} = c_1y_1 + c_2y_2$. By Exercise 2 $(\tilde{x}, \tilde{y})$ solves (1), and by Step 1 $(\tilde{x}(t_0), \tilde{y}(t_0)) = (x(t_0), y(t_0))$. Both (x, y) and $(\tilde{x}, \tilde{y})$ solve the same IVP at t_0, so by uniqueness $(x, y) = (\tilde{x}, \tilde{y})$ on $[a, b]$. ■

Where Exercise 3 enters

If the Wronskian were known not to vanish at only some t_0, the argument would work at that t_0; Exercise 3 shows that a non-zero value at one point already implies $W \neq 0$ everywhere, so one test is enough. Also, two solutions with $W \equiv 0$ cannot form a general solution: then the matrix above is singular at each instant, and not every initial vector can be matched.

INTERPRETATION AND CHECKS

- The solution space is a 2-dimensional vector space with basis $\{(x_1, y_1), (x_2, y_2)\}$ (a **fundamental set**; $\Phi = [\mathbf{x}_1\ \mathbf{x}_2]$ is a fundamental matrix and $\mathbf{c} = \Phi(t_0)^{-1}\mathbf{x}(t_0)$, Topic 4 Sec. 3).
- Example: for $\dot{x} = x + y$, $\dot{y} = 4x - 2y$ the solutions $e^{2t}(1, 1)$ and $e^{-3t}(1, -4)$ have $W = e^{2t}e^{-3t}(-4 - 1) = -5e^{-t} \neq 0$, hence form a fundamental set (Topic 4 Example 28.1).
- Check: if $x_2 = kx_1$ and $y_2 = ky_1$ (parallel solutions) then $W = kx_1y_1 - kx_1y_1 = 0$ and the argument rightly fails.
- The same proof works for $n \times n$ systems with $\det \Phi(t_0) \neq 0$.